

AYURVEDA

Ayurveda is a combination of two words: Ayu + Veda. Ayu means life and Veda means the science. Thus Ayurveda is a science of life and not only science of diseases

Āyurveda, meaning "the science of life," is an ancient holistic medical system that originated in India, emphasising balance and harmony between body, mind, and spirit.

According to Ayurveda, ayu i.e. life is a union of physical body, sense organs, mind and soul. The aim of Ayurveda is "Swasthasyarakshnam, aturasyavikaraprashaman" i.e. propagation of health and cure of diseases.

SAGE AGNIVESHA compiled the Agnivesha Samhita Sage Agnivesha was a legendary ancient Indian sage and one of the earliest authors on Ayurveda, a traditional Indian medical science.

A disciple of Sage Punarvasu Atreya, he compiled the teachings of his preceptor into a treatise called the [Agnivesha Samhita](#).

Key Aspects of Sage Agnivesha:

Disciple of Punarvasu Atreya: Agnivesha learned the science of Ayurveda from his teacher, Punarvasu Atreya.

Author of Agnivesha Samhita: He documented the knowledge he received in his own text, the Agnivesha Samhita.

Foundation of Charaka Samhita: The Charaka Samhita, a highly significant and enduring Ayurvedic text, was developed from Agnivesha's work.

Founder of an Ayurvedic School: He established one of the six early schools of Ayurvedic thought, known as the Agnivesha school.

Associated with Agni: Legends surrounding his birth link his descent or parentage to Agni, the god of fire.

Teacher of warfare: In some texts, like the Mahabharata, he is mentioned as the preceptor of Drona and Drupada in the art of warfare and weapons.

SAGE CHARAKA was a physician, he compiled the Charaka Samhita

- Sage Charaka was an ancient Indian sage, often called the "Father of Indian Medicine" and a key contributor to Ayurveda, who compiled the influential treatise Charaka Samhita.
- His work detailed aspects of physiology, diagnosis, and treatment, emphasizing holistic health and the importance of lifestyle and diet in preventing illness.
- He is credited with conceptualizing the three bodily principles (doshas)—Vata, Pitta, and Kapha—and his work established a foundational text for Ayurvedic practice for centuries
- Key Contributions
- Charaka Samhita: Charaka is best known for editing and expanding the ancient medical treatise Charaka Samhita. This work became a standard text for Ayurvedic medicine and was translated into many languages.
- Holistic Health: Charaka viewed Ayurveda as a holistic system, encompassing both preventative and curative care.
- Three Doshas: He elaborated on the concept of the three humors, or doshas—Vata (movement), Pitta (transformation), and Kapha (lubrication and stability)—which are fundamental to Ayurvedic understanding of health and disease.
- Prevention and Lifestyle: A central tenet of his philosophy was that prevention of disease through lifestyle, diet, and living in harmony with nature is more important than treatment.
- Medical Science: Charaka made significant contributions to fields such as physiology, anatomy, embryology, metabolism, and digestion.

SAGE VAGBHAT was a disciple of Sage Charaka, made significant contributions to Ayurveda

- Sage Vagbhat (or Vāgbhaṭa) was an influential classical Indian Ayurvedic physician and compiler of foundational medical texts, primarily known for authoring the [Ashtāṅgasaṅgraha](#) and [Ashtāṅghridayasamhitā](#).
- These influential works, part of the [Brihatrayees](#) (Great Trilogy) of Ayurveda, are considered cornerstones of traditional Indian medicine, outlining principles of human health, disease, and treatment.
- Key Contributions
- **Authoritative Texts:** Vagbhat wrote Ashtāṅgasaṅgraha (Comprehensive Collection) and Ashtāṅghridayasamhitā (Heart of the Eight Branches), which are comprehensive treatises on Ayurveda.
- **Foundational Knowledge:** These texts detail the understanding of the three fundamental energies of the body (Vata, Pitta, and Kapha) and their impact on health and disease.
- **Holistic Approach:** The works cover various aspects of human health, including diagnosis, treatment, and lifestyle principles that guide individuals from conception to death.
- **Influence:** Vagbhat's writings are revered for their refined style, scientific accuracy, and ability to concisely summarize and expand upon earlier Ayurvedic teachings.

SAGE SUSHRUTA was a surgeon who compiled the Sushruta Samhita

Which includes Plastic surgery, Neurosurgery, Orthopaedics, Pediatrics, etc

Sage Sushruta was an ancient Indian physician, surgeon, and teacher, considered the "Father of Surgery" and "Father of Plastic Surgery". His comprehensive surgical treatise, the [Sushruta Samhita](#), documents ancient Indian medical knowledge, outlining 300 surgical procedures, 120 instruments, and foundational techniques such as rhinoplasty. His teachings included anatomy through dissection, anesthesia, plastic and cosmetic surgery, and a rigorous ethical code for surgeons.

Key Contributions

Sushruta Samhita: This ancient treatise is a foundational text in Indian medicine and a major source of knowledge on ancient Indian surgery.

Plastic Surgery: Sushruta developed the "Indian flap" technique for nasal reconstruction (rhinoplasty).

Surgical Procedures: He detailed over 300 surgical procedures, including cataract surgery and the management of fractures and dislocations.

Surgical Instruments: The Sushruta Samhita describes more than 100 surgical instruments.

Anesthesia: Sushruta described methods for reducing pain during surgery, including the use of substances that acted as anesthetics.

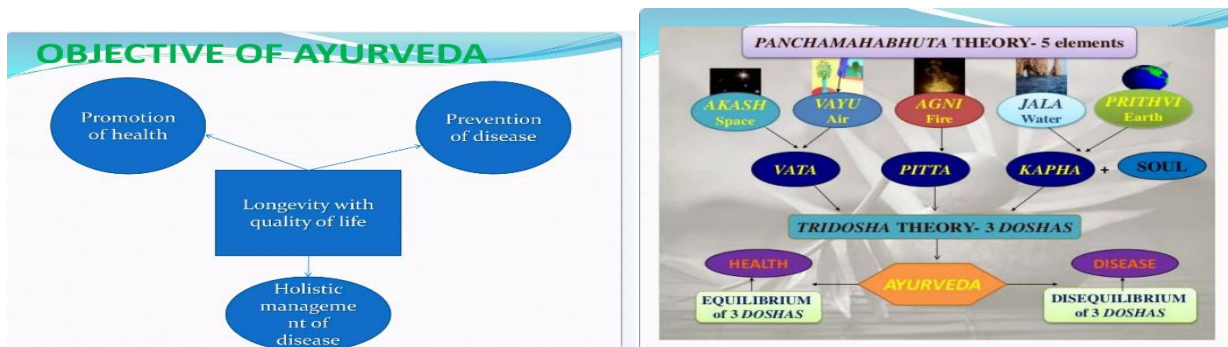
Ethical Principles: He established ethical guidelines and a code of conduct for surgeons, emphasizing practical training through simulation.

Anatomy: Sushruta advanced the study of human anatomy through dissection of human corpses to understand the inner structures of the body

OBJECTIVE OF AYURVEDA

The two principles objectives of Ayurveda are:

1. To prolong life and promote perfect health
 2. To completely eradicate (to destroy) the disease and dysfunction (not working normally) of the body
- To achieve "NIRVANA" OR Liberation from all kinds of wants



BASIC PRINCIPLES OF AYURVEDA

- Five gross elements (PanchaMahaBhutas)
 - Three biological energies (ThreeDoshas)
 - Three psychological energies (ThreeGunas)
 - Seven body tissues (SaptaDhatus)
 - Final refined product of digestion & metabolism, i.e., Vital Immunity (Ojas)
-
- According to ancient Indian philosophy, **the universe is composed of five basic elements or poncho bhutas:**
 1. **PRITHVI (EARTH),**
 2. **JAL (WATER),**
 3. **TEJA (FIRE),**
 4. **VAYU (AIR) AND**
 5. **AKASH (SPACE).**
 - **Everything in the universe, including food and the bodies were derived from these bhutas.**
 - **Dosha(biological energies found throughout the human body) in the body are comparable to universe**
 - **The human body is composed of derivaties of these five basic elements, in the form of**
 1. **TRI-DOSHA**

2. **TISSUES(DHATUS)**
3. **WASTE PRODUCTS(MALAS)**
- **The PANCHAMAHA BHUTAS, therefore serve as the foundation of all diagnosis and treatment modalities in Ayurveda**

BRANCHES OF AYURVEDA

THE EIGHT (ASHTANGA AYURVEDA)

1. Internal Medicine (Kayachikitsa)
2. Surgery (Shalyatantra)
3. Ophthalmology & Otorhinolaryngology (Shalakya Tantra)
4. Pediatrics (Kaumarabhritya)
5. Toxicology (Agadatantra)
6. Psychology (Bhutavidya)
7. Rejuvenation (Rasayana)
8. Revitalization (Vajikarana)

The Eight Branches of Ayurveda, also known as "Ashtanga Ayurveda," form the foundational pillars of this ancient holistic healing system. These branches were traditionally attributed to Dhanvantari, who is revered as the god of medicine in Hindu mythology and is believed to be the divine physician who first transmitted the knowledge of Ayurveda to humanity.

1. Kaya Chikitsa (Internal Medicine)

- **Focus:** This branch deals with the diagnosis and treatment of various diseases affecting the body. It emphasizes the balance of the three doshas (Vata, Pitta, and Kapha), as well as Agni (digestive fire) and the proper functioning of the dhatus (tissues).
- **Practices:** Includes dietary regulations, herbal remedies, detoxification methods (Panchakarma), and lifestyle modifications to maintain health and treat diseases.

2. Shalya Tantra (Surgery)

- **Focus:** Shalya Tantra is the branch of Ayurveda that deals with surgical techniques and procedures. It includes the treatment of physical injuries, wounds, fractures, and the removal of foreign bodies.
- **Practices:** Ancient texts describe various surgical instruments and procedures, including techniques for treating ulcers, tumors, and even cosmetic surgery. Sushruta, a renowned ancient surgeon, further developed and documented this branch.

3. Shalakya Tantra (Ophthalmology and ENT)

- **Focus:** This branch specializes in diseases and treatments related to the organs above the neck, primarily focusing on the eyes, ears, nose, throat, and head.
- **Practices:** It covers treatments for eye diseases, ear infections, nasal issues, and oral health problems. The use of herbs, oils, surgical tools, and therapies like Nasya (nasal therapy) is common.

4. Kaumarabhritya (Pediatrics and Obstetrics)

- **Focus:** Kaumarabhritya is concerned with prenatal care, childbirth, and pediatric care, including the growth and development of children.
- **Practices:** It includes guidelines for the health of the mother during pregnancy, the care of newborns, and the treatment of childhood diseases. This branch also covers immunization (Swarna Prashana), diet, and lifestyle for children.

5. Agada Tantra (Toxicology)

- **Focus:** Agada Tantra deals with the diagnosis and treatment of poisoning from various sources, including toxins from plants, animals, minerals, and environmental pollutants.
- **Practices:** It involves the use of antidotes, detoxification methods, and preventive measures to counteract the effects of poisons and venomous bites or stings. This branch also addresses poisoning due to food and water contamination.

6. Bhuta Vidya (Psychiatry)

- **Focus:** This branch addresses mental health and psychological disorders. It involves the study and treatment of mental imbalances, believed to be caused by external entities, negative energies, or imbalances in the mind-body system.
- **Practices:** Treatments include herbal formulations, mantra chanting, meditation, and therapies to calm the mind and balance the doshas. Bhuta Vidya also explores the use of rituals and spiritual practices to address mental health issues.

7. Rasayana (Rejuvenation and Anti-aging)

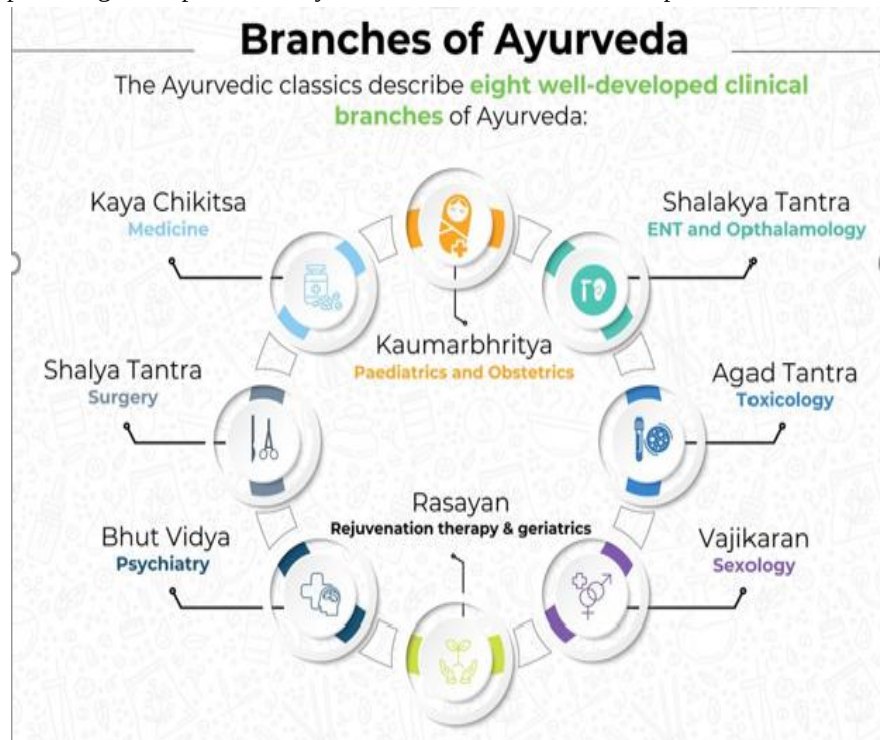
- **Focus:** Rasayana therapy aims at promoting longevity, vitality, and overall well-being by rejuvenating the body and mind. It is also concerned with boosting immunity, improving strength, and enhancing cognitive functions.

- **Practices:** The use of specific herbs, dietary regimens, and lifestyle practices are central to Rasayana. Chyawanprash and Ashwagandha are examples of Rasayana formulations. This branch also emphasizes the importance of a balanced lifestyle and mental clarity.

8. Vajikarana (Aphrodisiac Therapy and Reproductive Health)

- **Focus:** Vajikarana is the branch that deals with sexual health, fertility, and reproduction. It aims to enhance reproductive function, increase virility, and address infertility issues.
- **Practices:** This branch utilizes herbs, dietary practices, and lifestyle adjustments to improve sexual health and treat conditions like impotence and infertility. Vajikarana also emphasizes the importance of mental and physical harmony in sexual well-being.

These eight branches collectively address every aspect of human health, from preventive care and rejuvenation to the treatment of complex diseases and surgical interventions. The holistic approach of Ayurveda ensures that each branch is interconnected, providing a comprehensive system of medicine that has been practiced for thousands of years.



AYURVEDA BODY MIND RELATION

According to Ayurveda, ayu i.e. life is a union of physical body, sense organs, mind and soul. It advocates the measures for propagation of health and making the life of individuals beneficial to the society. It also describes the factors which may lead to pathogenesis in physical body as well as mind.

In Ayurveda, the body and mind are considered closely connected and interdependent. Health is achieved when the body, mind, senses, and soul remain in balance.

Basic Concept

Ayurveda believes that:

- Physical health affects mental health.
- Mental stress affects the body.
- Disease may originate in either the body or the mind and then influence the other.

Components of Human Existence in Ayurveda

Ayurveda explains human life through four major components:

1. **Sharira (Body)** – Physical structure and organs
2. **Manas (Mind)** – Thoughts, emotions, and mental activities
3. **Indriyas (Senses)** – Sense organs and perception
4. **Atma (Soul/Consciousness)** – Spiritual aspect of life

Harmony among these components leads to good health.

Tridosha and Mind-Body Relation

Ayurveda describes three biological energies called **Doshas**:

1. Vata Dosha

- Controls movement, nervous system, and communication.
- Imbalance causes fear, anxiety, restlessness.

2. Pitta Dosha

- Controls digestion, metabolism, and intelligence.
- Imbalance causes anger, irritability, stress.

3. Kapha Dosha

- Controls stability, immunity, and emotions.
- Imbalance causes laziness, attachment, depression.

These doshas influence both physical and mental functions.

Mental Qualities (Triguna Theory)

Ayurveda also explains the mind through three qualities called **Gunas**:

1. Sattva

- Purity, calmness, wisdom, positivity.
- Indicates healthy mental condition.

2. Rajas

- Activity, passion, excitement.
- Excess leads to anger and agitation.

3. Tamas

- Inertia, ignorance, dullness.
- Excess causes depression and confusion.

A healthy mind mainly possesses **Sattva guna**.

Factors Affecting Body–Mind Balance

- Food and nutrition
- Sleep
- Lifestyle habits
- Emotions and stress
- Environment
- Exercise and yoga
- Meditation and spiritual practices

Ayurveda emphasizes **proper diet, daily routine, yoga, pranayama, and meditation** for maintaining body–mind harmony.

Importance of Body–Mind Relation in Ayurveda

- Prevents diseases
- Improves mental peace
- Enhances immunity and energy
- Promotes emotional stability
- Supports long and healthy life

KOSHA

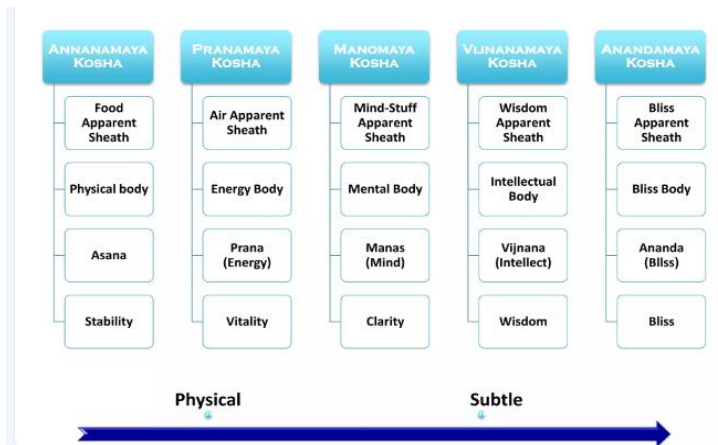
Meaning "sheath" or "covering" in Sanskrit. A kosha usually rendered "sheath", is a covering of the Atman, or Self according to Vedantic philosophy.

The **koshas** are considered the energetic layers of your body that surround your soul. Sometimes, they're referred to as "sheaths" or "casings."

Consciousness in humans flows in 5 layers of the body to reach the Atman (or Self, according to Vedanta philosophy). These 5 layers in yogic tradition are called koshas

The ancient concept of "Personality" or "Body constitution", underlines the below mentioned 'five' sheaths (an investing cover).





ANNAMAYA KOSHA

- First layer – seen easily, is visible, can be controlled easily
- CONSISTS OF - Physical Body, All the Physiological Systems
- ‘Annamaya’ [Anna means food and Maya means ‘made of’] is a segment of the human system is nourished by food (anna).
- Based on ‘food’ consumption – (□□□□) Hence the Name Annamaya Kosha
- Here “food” basically not what is generally understood , Includes all source of energy – Air + Water + Food
- Here food represents the ‘Physical matter’ from which the body formation took place (5 elements)
- This sheath contains the 5 elements (earth, space, air, water, fire) that constitute the physical body with a dominance of ‘Tamoguna’.
- It’s perishable in nature and hence, has a beginning and an end (birth and death).
- It’s the reason, the first layer of the body is linked with the Root chakra & earth element present in our body.

PRANAMAYA KOSHA

- Prana is the life force that is present in the entire cosmos
- Pranamaya’ (vital air sheath) segment is nourished by ‘Prana’.
- Second Layer – just outside of the Annamaya Kosha
- 1. CONSISTS OfPranic Energy – hence the name
- 2. Nadi’s
- 3. Chakras
- This sheath contains the 5 pranas that manifest in the physical body and connecting it to the next kosha i.e. Manomaya kosha.
- This sheath has a dominance of Rajoguna, hence subtle activities are normal in this layer.
- This sheath is also perishable and has a beginning and an end.
- In yoga, both asanas and pranayama (breathing exercises) affect this kosha.

MANOMAYA KOSHA

- Third Level – Mental Level
- Refers to the Mind
- Mind does not have any biological structure.
- Mind does not have any physical dimension
- Manomaya’ (mental sheath) is the segment nourished by knowledge.
- Yet Mind is most important part ,It has capacity to move outwards and understand the outside world
- Also capacity to move inwards – experience & understand Inner Self
- Dual function ; -Seat of emotions & thought process & patterns
- This sheath contains gyanendriyas&karmendriyas for interaction with the outer world.
- 1. Gyanendriyas are sense organs through which one perceives objects of the world and
- 2. karmendriyas are organs of actions through which one manipulates with the objects of the world
- The primary functions of manomaya kosha are:
- 1. *Sankalpas* i.e. being the aspects to interpret the intention and act accordingly.

2. *Vikalpas* refers to rejecting undesirable actions mostly with negative outcome.
 - This kosha is made of a combination of Sattva guna and Tamoguna.
 - It's also perishable in nature and has a beginning and an end.

VIJNANAMAYA KOSHA

- Fourth Layer – just outside of Manomaya Kosha
- Works at the Intellect Level / Intelligence Level , Hence the Name – Vignanamaya Kosha
- Vijnana literally means intellect, hence Vijnanamaya kosha is the intellect/wisdom/knowledge sheath.
- Vijnanamaya' (intellectual sheath) is nourished by 'ego'.
- this kosha is related to the throat chakra & predominant with the air element.
- Has the ability to discriminate or discern
- Ability to understand, discriminate, analyse judge – correctly
- Seat of correct thought process, proper judgement proper rational thinking & correct decision
- It's composed of a combination of intellect and the five sensory organs and is also considered to be the part of one's being that is responsible for will, discernment, and determination.

ANANDAMAYA KOSHA

- 5TH & Final Level of Existence
 - Most subtle of all
 - Ananda means blissful experience.
 - In Advaita Vedanta, anandamaya kosha is referred to as the innermost Kosha having close proximity with the soul, hence experiences the blissful experience coming out of the soul.
 - Anandamaya sheath marks the segment which is nourished by emotions and consciousness.
 - This is the intuitive expansive sheath aligned with the causal body and is often thought of as the soul (atman)
 - The anandamaya kosha highlights the three positive blissful qualities of the Soul viz Sat, Chit and Anand.
1. Sat: being truthful and eternal.
 2. Chit: it refers to the one which is alive and has the consciousness, the main bridging line separating the living and the non-living.
 3. Anand: it refers to an ever-joyful state.

TRI-DOSHA

Dosha(bio-energies) are structural and functional component of the human body. Though, the term 'Doṣa' means 'the disturbing factor', it has got definite physiological importance in normal state

In general dosha means bodily dosha, But doshas are two types –

1. bodily &
2. Psychological

Bodily doshas are 3 types - vata pitta kapha

psychological doshs are 2 types - rajas & tamas

Basically three Doshas (Vata Pitta and Kapha)- are responsible for maintenance of homeostasis in the body, and health is nothing but a state of equilibrium of these tri dosha. Disease is manifested as a result of disturbance in the state of equilibrium among these Doshas. The state of equilibrium among these „Doshas“ is responsible for maintenance of health.. The concept of „Tridosha“ is basically a theory and any single substance or structure in the body can not represent a „Dosha

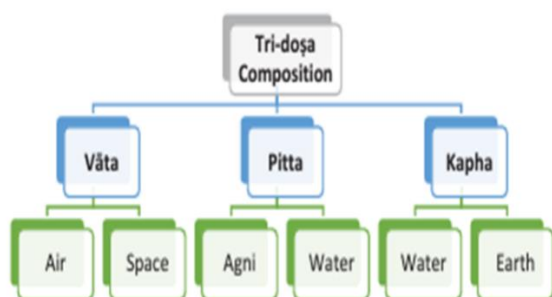


FIGURE 13.2 The Composition of the Tri-Doṣas

VATA - Air & Space

- VATA QUALITIES
1. Dry, rough

2. Cold
3. Light
4. Subtle, minute
5. Changeable, moving
6. Clear, non-sticky
7. Coarse, brittle
 - Vata season is autumn(season between summer and winter)
 - Time of day is afternoon and early morning

VATA -TYPES & THEIR FUNCTIONS

1. Prana(a life-giving force)Vata : Governs the senses, creative thinking, reasoning, enthusiasm,
2. Udana ("to soar" or "to fly")Vata : Governs quality of voice, memory, movements of thought
3. Samana('equal' or 'same'.) Vata : Governs movement of food through digestive tract
4. Apana(The outgoing)Vata : Elimination of wastes, sexual function, menstrual cycle
5. Vyana("outward-moving air")Vata : Governs blood flow, heart rhythm, perspiration, sense of touch

VATA is responsible for:

Vata governs the principle of movement and therefore can be seen as the force which directs nerve impulses, circulation, respiration and elimination, etc

1. Movement,
2. Transportation
3. Communication ,
4. Neurological activities

FUNCTIONS OF VATA

1. **Breathing**
2. **Cellular motion**
3. **Flow of nutrients & waste products**
4. **Heartbeat, blood flow**
5. **Movement**
6. **Natural urges**
7. **Nerve impulses, blinking**
8. **Sensory & motor** functions

LOCATIONS OF VATA

- Colon (Main)
- Bladder / kidneys
- Anus & rectum
- Waist
- Pelvis,
- reproductive organs
- Legs,
- feet Bones
- Ears
- Skin

BALANCED FUNCTIONS OF VATA

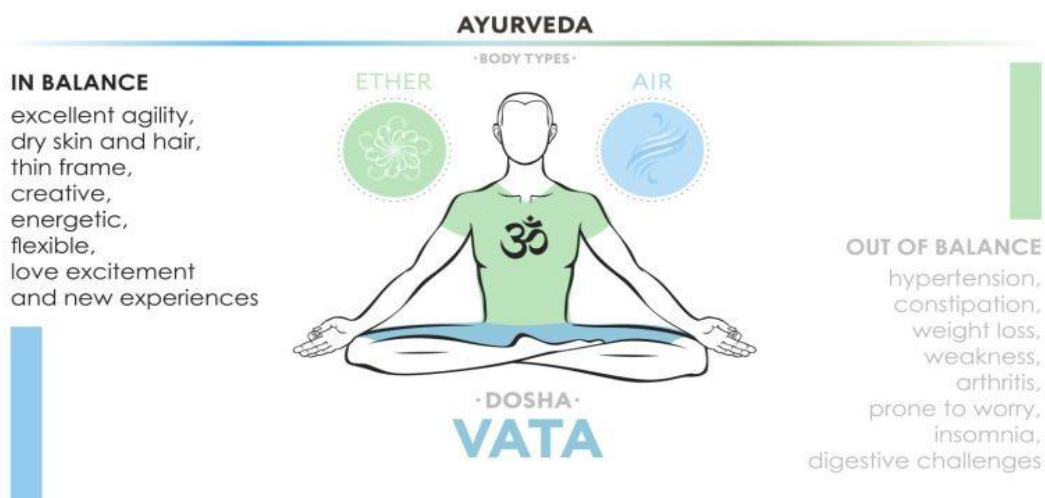
PHYSICAL

1. All movements of the body
2. Circulation of the blood
3. Control of the organs of action and perception
4. Curiosity and natural interest
5. Development of tissues
6. Excellent energy level
7. . Flow of nutrients
8. Greater vitality
9. Liveliness
10. Motivation of natural urges

11. Nerve impulse, pulsation, heartbeat
12. Normal respiratory function
13. . Proper coordination of all body functions
14. Release of waste products
15. Sensory & motor functions
16. Sound sleep
17. Stimulation of digestive juices
18. Strong immunity

PSYCHOLOGICAL

1. Clear comprehension
2. Controlled and precise mental activity
3. Creativity
4. Desire to lead an active life
5. Enthusiasm
6. Inspiration
7. Mental alertness
8. Sense of exhilaration



PITTA

PITTA - Fire & Water

PITTA QUALITIES

1. Slightly oily
2. Heat, warmth
3. Sharp
4. Liquid
5. Sour
6. Moving, flowing
7. Pungent, sour
8. spreading qualities.

TYPES & THEIR FUNCTIONS

1. Alochaka ("critic") Pitta : Controls functioning of the eyes
2. Bhrajaka (that which makes things bright')Pitta : Responsible for healthy glow of the skin
3. Sadhaka(any religious practitioner.) Pitta : Controls desire, drive, decisiveness, spirituality
4. Pachaka (Digestive)Pitta : Responsible for digestion, assimilation, metabolism for healthy nutrients and tissues
5. Ranjaka ('to give colour,)Pitta : Responsible for healthy, toxin-free blood
 - Responsible for the process of transformation or metabolism.
 - The transformation of foods into nutrients that our bodies can assimilate is an example of a Pitta function.
 - PITTA is responsible for:
 1. Digestion
 2. Metabolism

3. Transformation
4. Hormonal,
5. Enzymatic actions
6. **Absorption**
7. **Assimilation**
8. **Body temperature**
9. **Perception**
10. **Skin coloration & luster**
11. **Understanding**
12. **Vitality**

LOCATIONS OF PITTA

1. Small intestine and stomach (Main)
2. Sweat
3. Plasma, blood
4. Liver, spleen
5. Endocrine glands
6. Skin
7. Brain
8. Eyes

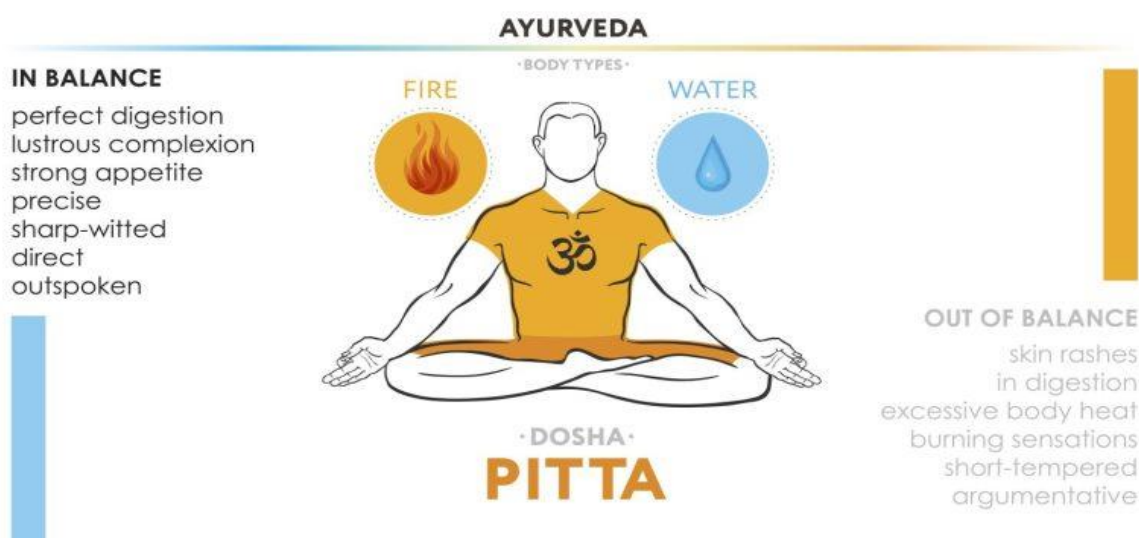
BALANCED FUNCTIONS OF PITTA

PHYSICAL

1. Good appetite
2. Balanced digestion
3. Balanced heat regulation
4. Balanced thirst
5. Good eye sight
6. Luster Softness in the body

PSYCHOLOGICAL

1. Cheerfulness
2. Exhilaration
3. Valor
4. Sharp
5. Intellect



KAPHA - Water & Earth

• KAPHA QUALITIES

1. Heavy
2. Cold
3. Soft

4. Unctuous, oily
5. Sweaty
6. Stable
7. Sticky
- Heavy, slow, cool, oily, liquid, dense, thick, static and cloudy qualities.

KAPHA - TYPES & THEIR FUNCTIONS

- Tarpaka (one that satisfy or nourishes") Kapha : Responsible for moisture for nose, mouth, eyes and brain
- Bhodaka (knowledge')Kapha : Governs Sense of taste, which is essential for good digestion
- Kledaka ("wetting" or "moistening".)Kapha : Controls moisture of the stomach and intestinal mucosal lining
- Avalambaka(support and dependence) Kapha : Protects the heart, strong muscles, healthy lungs
- Sleshaka ("binding or hugging")Kapha : Lubricates the joints, Keeps skin soft and supple.
- Kapha is responsible for growth, adding structure unit.
- It also offers protection, for example, in form of the cerebral-spinal fluid, which protects the brain and spinal column.
- The mucousal lining of the stomach is another example of the function of KaphaDosha protecting the tissues.
- KPHA is responsible for:
 1. Structure
 2. Cohesion
 3. Binding
 4. Lubrication
 5. Immune activities

LOCATIONS OF KAPHA

1. Thorax (Main)
2. Head, neck
3. Mouth
4. Joints
5. Kidneys
6. Adipose tissue
7. Lymphatic system
8. Stomach,
9. small intestine
10. Pancreas

BLANCED FUNCTIONS OF KAPHA

PHYSICAL

1. Binding, unctuousness
2. Heaviness
3. Lubrication of joints
4. Plumpness
5. Potency
6. Stability of the body Strength,
7. strong immunity
8. Vitality
9. Wound healing

PSYCHOLOGICAL

1. Absence of greed
2. Affection
3. Enthusiasm
4. Firmness
5. Forbearance and fortitude
6. Forgiveness Generosity, knowledge
7. Restraint, patience
8. Understanding

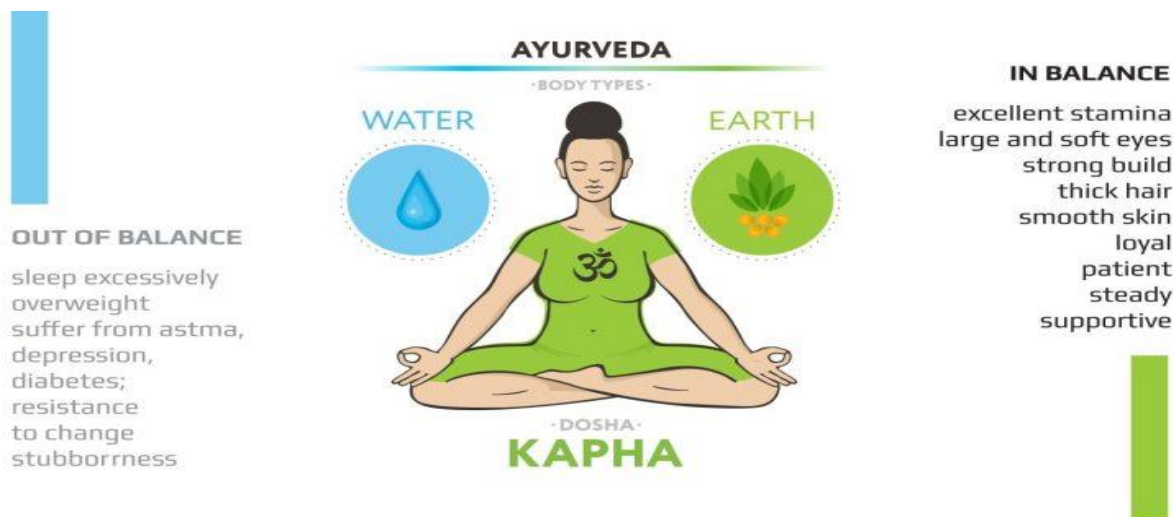


TABLE 13.3 The Effects of Doṣas Going Out of Balance

Condition	Vāta	Pitta	Kapha
Balanced	Energy, balanced breath, movement, normality in urges	Digestion, warmth, vision, hunger, thirst, taste, gleam, intelligence, prowess/heroism, delicateness, softness	Stability, moisture, good joints, patience
Exaggerated	Thinness, need of warmth, shivering, constipation, weakness, chattering, dizziness	Yellowness in eyes and excretions, severe hunger and thirst, burning, insomnia, acidity	Indigestion, nausea, laziness, heaviness, frigidity, slack-limbed
Reduced	Exhaustion of limbs, noxious talk, giddiness	Indigestion, coldness, lack of gleam	Giddiness, emptiness in the abode of kapha, loose joints

DRAVYA GUNA KARMA AYURVEDA

Dravya, guna, and karma are concepts in Ayurveda that are used to describe the nature, qualities, and pharmacological activities of herbs and Ayurvedic medications:

Dravya: The substance or drug

Dravya(Drug): A substance which possess Guna (property) and Karma (action) with inheritance relation is known as Dravya

The subject of dravyaguna deals with the study of drugs, including herbs, plants, and metallic compounds

Guna: The property or quality of the substance

Guna: The qualities of medications used in Ayurveda.

Guna (Physical property) – It is the property which will have inherent reaction with dravya but remain inactive

Karma: The action or effect of the substance

Karma: The word Karma refers to by which the action will be take place.

It is responsible for Samyoga (Combination), Vibhaga (Division) and is resides in Dravya

DRAVAYA

The Sanskrit word ‘dravya’ is derived from ‘dru-gatau’ dhatu (verb) having ‘yat’ pratyaya (suffix).

Charak : that which is substratum of action (Karma) and Qualities(Guna) and is the co-existent cause, is known as Dravya (substance).

It means anything which has movement and attainment or which changes to resultant.

Dravya literally translates to 'substance,' and is used to distill Ayurvedic medicines.

This pharmaceutical substance is composed of the Five Great Elements or the Mahabhutas, and contains their specific qualities within it.

- The two main goal of Ayurveda are
 1. To protect health of a healthy person
 2. To eradicate the diseases

This two goals are achieved with the help of dravya

In Ayurveda, Parthiva Dravya is a substance that has a predominance of Prithvi mahabhuta, or the element of earth:

TYPES OF DRAVYA...

1. Parthiva Dravya
2. Aapya Dravya
3. Aagney Dravya
4. Vayaviya Dravya
5. Aakashiya Dravya

Properties: It has properties

1. like being stout (sthool),
2. durable (sthira),
3. having a specific shape (murtimad),
4. food taking long time for digestion (guru),
5. rough (khar), and
6. hard (kathinam).

Examples :-Nails, tendons, bones, teeth, mamsa, skin, hair on head, hair on body, beard, nose, and osmotic apparatus are all Parthiva Dravya

Āpyadravya (आप्यद्रव्य) are those substances (dravya) related to Jala-mahābhūta obtained from the Āpya type of “land” (kṣetra),

- Āpyadravya (आप्यद्रव्य) are those substances (dravya) related to Jala-mahābhūta obtained from the Āpya type of “land” (kṣetra),

Agneya dravyas are substances that have the following qualities:

1. Ruskha: Dry
 2. Tikshna: Sharp
 3. Usna: Hot in potency
 4. Visada: Clear
 5. Sukshma: Subtle
 6. Rupa guna: Predominance of rupa guna
- Agneya dravyas are responsible for the following:
 1. daha,
 2. prabha,
 3. varna,
 4. prakasha, and
 5. pachana

Vāyavyadravya (वायव्यद्रव्य) are those substances (dravya) related to Vāyu-mahābhūta obtained from the Vāyavīya type of “land” (kṣetra), ...

- Vayaviya dravya are substances that have specific properties,
- such as being light,
- dry,
- cool,
- rough,
- clean, and
- minute.
- They are characterized by the touch guna (sparsh).
- Some of the functions of vayaviya dravya include:
 1. Causing dryness,
 2. Causing lightness,
 3. Causing clarity,
 4. Causing tiresomeness, and
 5. Sustaining the dhatus

Ākāśyadravya (आकाशीयद्रव्य):—Substances having akāśa as the predominant mahabhuta;

- Akashiya Dravya-Substance which have the properties
 1. like soft,
 2. lighter,
 3. minute,
 4. glesening,
 5. clear,
 6. sepearte,
 7. unknown taste etc.
- The substance produce smoothness of the body

GUNA

Represents the quality of a Dravya, and is considered the working capacity.

There are many types of Guna, including

1. Laghu;-lightness' quality.
2. Tikshna;- quality of sharpness and penetration.
3. Ushna, ;- 'hot / heat' quality.
4. Snigdha;- "smooth and greasy or "oily".
5. Ruksha;- "dry" or "dryness".
6. Khara;- property which makes the surface rough
7. Sandra;- means dense.
8. Kathina;- "hard" or "stiff"
9. Sthira;- 'immobility or stable'
10. Sara;- 'mobility' quality
11. Sukshma;- "fineness" or "subtle" quality.
12. Sthula;- 'bulkiness / stoutness' quality.
13. Vishada;- 'clearness' quality
14. Picchila;- 'sliminess' quality

KARMA

- The factor responsible for the action of drug is Karma
- The inherent relation factor in drug which is responsible for association (Samyog) and disassociation (Viyog) is termed as Karma.
- According to Bhadant Nagarjuna Action/ therapeutic action that is seen is karma eg- vaman, virechan etc

Relationship Between Dravya, Guna and Karma

Dravya → possesses → Guna

Guna → produces → Karma

Thus:

- Substance is the base
- Qualities determine nature
- Actions produce therapeutic effects

Importance in Ayurveda

These principles are used for:

- Selection of medicines

- Diet planning
- Dosha balancing
- Disease treatment
- Understanding drug action

DISEASE MANAGEMENT IN AYURVEDA

- **Caraka-sarphita outlines four components of a disease management scheme:**
 1. **THE PHYSICIAN,**
 2. **THE DRUG,**
 3. **THE PATIENT, AND**
 4. **THE ATTENDANT'S.**
 - The physician comes first in order of importance.
 - She/He must possess technical skill, scientific knowledge, purity and human understanding.
 - The physician should use his/her knowledge with humility, wisdom and in the service of humanity.
 - A physician must have proper training, knowledge, and experience.
 - Next in importance comes food and drugs.
 - These are supposed to be of high quality, wide application, grown and prepared following approved procedures and should be available adequately.
 - A remedy must be abundantly available, effective, and relatively **safe**.
- **THE ATTENDANT'S.**
 - **The third component of every successful treatment is the role of nursing personnel who should have good knowledge of nursing, must know the skills of their art and be affectionate, sympathetic, intelligent, neat & clean and resourceful.**
- **THE PATIENT**
 - **The fourth component is the patient himself who should be cooperative and obedient to follow instructions of the physician, able to describe ailments and ready to provide all that may be needed for treatment.**
 - **A patient must provide all information to the physician about the disorder and be compliant and an attendant must have the knowledge of patient care and cleanliness.**

Steps

- The treatment framework in Ayurveda consists of three basic steps:
 1. Symptoms (Lingo),
 2. Cause (Hetu), and
 3. Remedy (Aupdha).

Symptoms :-The process of curing a disease invariably begins with the recognition of symptoms.

- Symptoms merely provide the starting point for understanding the cause.
- Ayurveda differentiates itself from the modern schools of medicine in the manner in which the symptoms are further pursued to identify the root causes.

HETU:- Basically the meaning of Hetu is Kaarana or cause.

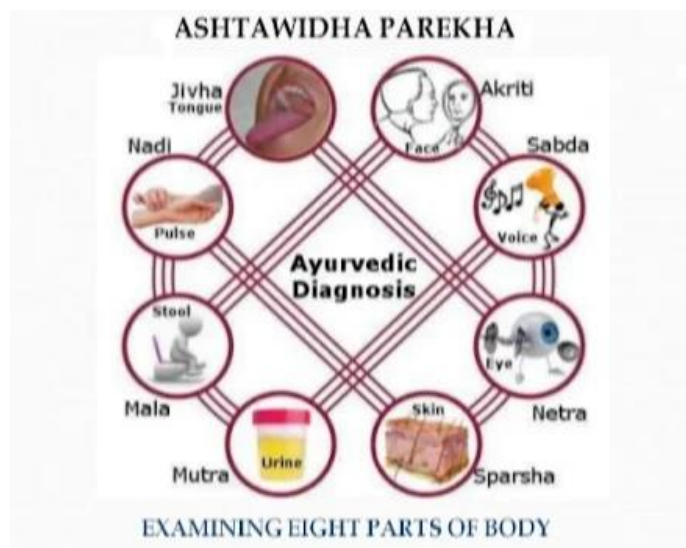
- Madhavakar has defined Hetu as, 'The cause which develops disease after going through different processes
- According to Acharaya Charaka 'The cause which lights the knowledge or through which the true knowledge is gained' is Hetu
- In other context Chakarapani define it as 'The cause of developing disease is Hetu

Diagnostic Techniques

- Ayurveda lays great emphasis on an early and correct diagnosis.
- These medical techniques can be categorised into three groups:
 1. Diagnostic Techniques,
 2. Prognostic Techniques and
 3. Therapeutic Techniques.
- Diagnostic tools: These are tools aimed at predicting, diagnosing, or grading specific disorders enumerated in classical textbooks;
- Prognostic tool: Ayurvedic literature is abundant with signs and symptoms which indicate good as well as bad prognosis, which in turn, can be used to make predictions regarding the probable therapeutic outcomes.
- It is based on the main fundamental principles of the five elements, the three doshas, the four stages of life, the six tastes and the concept of Agni. By understanding these fundamental principles of Ayurveda and applying them to our daily lives, we can achieve a balance between our mind, body and spirit

Examination of the Patient	Examination of the Disease
♦ Prakṛti (Constitution)	♦ Darśana-parīkṣā
♦ Vikṛti (Morbidity)	♦ Sparsana-parīkṣā
♦ Sāra (Tissue elements)	♦ Prasna-parīkṣā
♦ Saṃhanana (Organs)	♦ Nādi-parīkṣā
♦ Pramāṇa (Body measurements)	♦ Mūtra-parīkṣā
♦ Sātmya (Homologation)	♦ Mala-parīkṣā
♦ Sattva (Psychic condition)	
♦ Āhāra-śakti (Food intake)	
♦ Vyāyāma-śakti (Physical fitness)	
♦ Vayas (Age)	

FIGURE 13.3 Diagnostic Technique Prescribed in Āyurveda Texts



Darsana-pariksa:

- This is a visual technique of inspection of the patient, the affected parts etc. The colour, lusture, external appearance, size, shape, etc. are assessed.

Sparsana-pariksa:

- By a physical examination of the patient (palpation) the physician can get knowledge of qualities such as heat, pulsations, etc.

Prasna-pariksa:

- Engaging the patient (and the well-wishers and accompanying people if the patient is unconscious) is a technique to elicit some information which will be helpful in the diagnosis and prognosis of a disease.

Nadi-pariksa:

- This is a common practice adopted by an Ayurvedic practitioner although there is no description of this in the earlier texts. It ought to have been introduced during the medieval period. By observing the pulse, the health of the individual is assessed.

mutra-pariksa:

- In addition to the usual urine examination, Ayurveda also prescribes an oil drop test (Taila-bhmlu-parik0) in which a drop of oil is left on the surface of the urine and observed. The movement of the oil drop in different directions and its rate of spread and change in colour are used to assess a number of chemical and physical properties.

Mala-pariksa

- Examination of faeces is a technique for diagnosis of a disease. In Ayurveda, an analysis of the movement of the faecal matter when dropped in water is made to assess the digestive and metabolic conditions of the patient

Types of Treatment

- The treatment of disease can broadly be classified as

Shodhana therapy (Purification Treatment)

- Shodhana treatment aims at removal of the causative factors of somatic and psychosomatic diseases.
- The process involves internal and external purification.
- The usual practices involved are Panchkarma (medically induced Emesis, Purgation, Oil Enema, Decoction enema and Nasal administration of medicines), Pre-panchkarma procedures (external and internal oleation and induced sweating).

Shamana therapy (Palliative Treatment)

- Shamana therapy involves suppression of vitiated humours (doshas).
- The process by which disturbed humour subsides or returns to normal without creating imbalance of other humours is known as shamana.
- This treatment is achieved by use of appetisers, digestives, exercise and exposure to sun, fresh air etc. In this form of treatment, palliatives and sedatives are used.

Pathya Vyavastha (Prescription of diet and activity)

- Pathya Vyavastha comprises indications and contraindications in respect of diet, activity, habits and emotional status.
- This is done with a view to enhance the effects of therapeutic measures and to impede the pathogenetic processes.
- Emphasis on actions such as dos and donts of diet is laid with the aim to stimulate Agni and optimize digestion and assimilation of food in order to ensure strength of tissues.

Nidan Parivarjan (Avoidance of disease causing and aggravating factors)

- Nidan Parivarjan is to avoid the known disease causing factors in diet and lifestyle of the patient.

- It also encompasses the idea to refrain from precipitating or aggravating factors of the disease.

Satvavajaya (Psychotherapy)

- Satvavajaya concerns mainly with the area of mental disturbances.
- This includes restraining the mind from desires for unwholesome objects and cultivation of courage, memory and concentration.

Rasayana therapy (use of immunomodulators and rejuvenation medicines)

- Rasayana therapy deals with promotion of strength and vitality.
- The integrity of body matrix, promotion of memory, intelligence, immunity against the disease, the preservation of youth, lustre and complexion and maintenance of optimum strength of the body and senses are some of the positive benefits credited to this treatment.
- Prevention of premature wear and tear of body tissues and promotion of total health content of an individual are the roles that Rasayana therapy plays.

ALLOPATHY	AYURVEDA
means sickness together it means 'Other than illness'	Ayurveda 'Intelligence or Knowledge of Life'
the approach to Allopathy is not holistic.	The approach to Ayurveda is more holistic
Allopathy is significantly higher in cost than Ayurveda	Ayurveda is significantly lower in cost than Allopathy
Opts preventing course of action	Opts preventing curative course of action
Chemical Sources	Natural Sources
Instant Relief	Long Term Relief
Life Long Medication	Health Food – Life Style
Significant Side Effect	Negligible Side Effects
Works on symptoms only	Works on root cause of diseases
Diagnosis done by use of five sense organs	Diagnosis done by laboratory equipments and investigation

AYURVEDA MIND BODY RELATION

- in Ayurveda, the mind (*manas*) and body are fundamentally interconnected, influencing each other in a continuous, dynamic relationship.
- Health is defined as a state of balanced doshas, a strong digestive fire (*agni*), properly functioning tissues, and an overall sense of deep happiness in the mind, body, and spirit.
- Ayurveda is composed and designed with a set of logic and rules that when together implemented will have a focus on mind, body, and spirit for optimal health
- Ayurveda recognizes the connection between the mind and body and the impact that mental health has on physical health.
- Negative emotions, stress, and anxiety can lead to physical health problems, while positive emotions and a calm mind can promote well-being.
- Ayurveda uses a holistic approach to mental health, focusing on lifestyle practices and treatments that promote balance and harmony between the mind and body.
- Ayurveda emphasizes the profound connection between the mind and body. It acknowledges that emotional and mental well-being greatly influence physical health.
- Through practices like meditation, yoga, and mindfulness, individuals can cultivate a deeper awareness of their thoughts, emotions, and physical sensations, promoting holistic healing.

ASHTANGA YOGA is one of the classifications of Yoga in Patanjali's Yoga Sutras. **Ashtanga Yoga** is the eightfold path of yoga described by Patanjali in the **Yoga Sutras**.

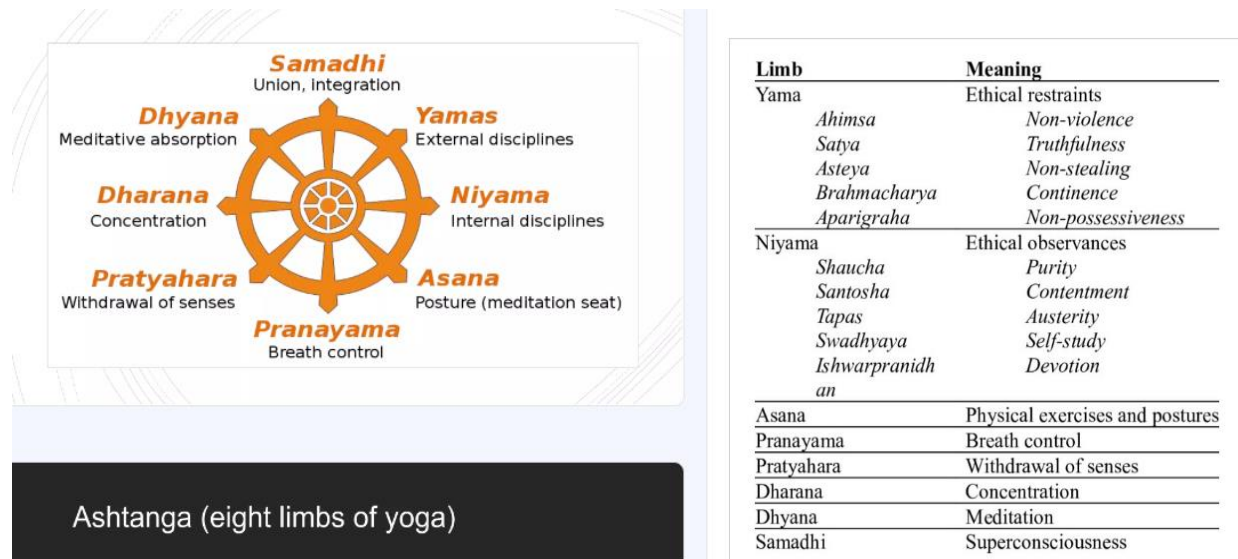
The word:

- **Ashta** = Eight
- **Anga** = Limbs or parts

It is a systematic method for achieving physical health, mental discipline, and spiritual growth.

Ashtanga yoga is a dynamic and vigorous style of yoga that synchronize breath with a specific , set sequence of posture

Ashtanga Yoga is where you move through different poses while matching your breath with each movement. The purpose of Ashtanga yoga is the purification of the body and mind to achieve a state of holistic well-being, mental clarity, and spiritual growth



- 1) YAMA(Principles) – Values and Ethical Practices such as being Honest / Truthful;
- (2) NIYAMA (Personal Discipline)– “Rule” – Moderate in needs, moderate in activity; Selfless work
- (3) ASANA – Posture – Physical Activity
- (4) PRANAYAMA (Breathing exercises / practices) – Control over Breathing and Control over “Prana”
- (5) PRATYAHARA (Isolation)- Withdrawal from reacting to the sense organs which stimulates the mind.
- (6) DHARANA (Concentration)– Controlling the Mind Vs Observing the Mind
- (7) DHYANA:(Meditation) Mind is absorbed in the Intellect; Only the SELF remain in this state.
Very often people call this Meditation – without doing Step 1 to 6, they want to directly come to Step 7...and PRETEND THAT THEY ARE DOING MEDITATION
- (8) SAMADHI(Salvation) – EGO (SELF) IS CONNECTED TO UNIVERSAL CONCIOUSNESS

1. Yama (Good Habits for Life) (Moral Discipline)

These are ethical rules for social behavior.

Five Yamas

Yama	Meaning
Ahimsa	Non-violence
Satya	Truthfulness
Asteya	Non-stealing
Brahmacharya	Control of senses/celibacy
Aparigraha	Non-possessiveness

Ahimsa: Path to Non-Violence

Satya: Path to truthfulness.

Asteya – Path to not stealing.

Brahmacharya – Path to self-focus and Celibacy

Aparigraha – Path to not being greedy

Importance

- Develops good character
- Promotes peace and harmony

2. Niyama (Good Personal Habits) (Personal Discipline)

Rules for self-discipline and personal conduct.

Shoucha – Path to cleanliness

Santosh – Path to Happiness

Tapa – Path to Endurance

Swadhyaya – Path for Self-growth and Self-Study

Eshwar Pranidhan – Path to Humbleness

Importance

- Improves self-control
- Builds inner purity

3. Asana (Yoga Poses) (Physical Posture)

Physical postures practiced to keep the body healthy and stable.

Sitting or standing in a way that makes your body at ease.

Yoga poses: asana

Benefits

- Improves flexibility
- Increases strength
- Enhances concentration
- Maintains physical fitness

Today, most people practicing yoga focus on the practice of the third limb, Asana, or yoga postures.

Yoga poses are a discipline of physical postures designed to purify the body and provide physical stability, stamina, and strength.

In ancient times, asana practice was meant more for cultivating the physical and mental stability and easiness that is required for the practice of meditation (the seventh limb). The literal meaning of asana is “a seat”.

Asanas include the dual qualities of alertness (steadiness) and relaxation (easiness).

Asanas give us an opportunity to study and explore the physical realm of our existence. More than flexibility of muscles and alignments, they are about connecting physical steadiness to stillness and happiness of mind.

Asanas involve increased awareness of various physical and physiological processes influenced by controlled stretching, contraction, relaxation, and coordination of various muscles, and in balancing and maintenance of posture

4. Pranayama (Breathing Exercises)

Learning to breathe in control.

Control and regulation of breathing.

Types

- Anulom Vilom
- Bhastrika
- Kapalabhati

Benefits

- Improves lung capacity
- Calms the mind
- Increases energy

Pranayama consists of two Sanskrit words—Prana (life force) and Ayama (control). So, pranayama means the regulation of the flow of life force.

It is the science of breathing. Breath is the physical expression of life-force energy. In recent years, the term breathwork has been coined to describe breathing exercises or breathing techniques.

We breathe every single minute. We take approximately 21, 600 breaths every day!

Breath is a bridge between our body, mind and spirit. Changes in our mind affect our breathing and vice-versa. Every emotion has a corresponding rhythm of the breath, and the way we breathe can influence the way we feel.

Breathing is also a bridge between our sympathetic and parasympathetic systems.

Practice of pranayama helps us to harness the power of breath for physical, mental, emotional and spiritual health.

5. Pratyahara (Controlling Your Senses)

Meditating to avoid distractions and to build up your senses.

Pratyahara (Withdrawal of Senses)

Control of senses from external objects.

Importance

- Reduces distractions
- Improves concentration
- Prepares mind for meditation

Most of the time we are engaged with the outer world through our senses. Our senses distract us from coming inwards. The various techniques of Pratyahara teach us to withdraw our senses from the outer world. These techniques allow the mind to settle inwards.

6. Dharana (Focusing Your Mind)

Paying full attention to one thing at a time.

When the mind settles inwards we can train it to focus effortlessly.

Dharana techniques help us in maintaining this inward attention. Get your earphones and try this Dharana to Dhyana Meditation. Fixing the mind on a single object or thought.

Examples

- Focusing on breath
- Concentrating on a mantra
- Candle meditation

Benefits

- Sharpens memory
- Develops mental stability

7. Dhyana (Meditation)

Sitting quietly and thinking deeply without distractions.

Meditation or Dhyana is the effortless art of doing nothing in order to experience inner peace.

It is the skill of relaxing the mind totally and living in the present moment.

Through regular practice of meditation, we can remain in witness consciousness and achieve a blissful state with complete evenness of mind and harmony with the universal consciousness

Continuous flow of concentration leading to meditation.

Benefits

Mental peace

Emotional balance

Spiritual awareness

8. Samadhi (Feeling Pure Happiness and Peace)

Reaching a state where you feel completely happy and feeling connected to everything.

Flow of Ashtanga Yoga; Yama → Niyama → Asana → Pranayama → Pratyahara → Dharana → Dhyana → Samadhi

BENEFITS OF ASHTANGA

Regular, dedicated practice is said to bring numerous benefits, including:

1. **Increased physical strength and flexibility: The challenging, dynamic sequences effectively build muscle strength and increase range of motion.**
2. **Detoxification: The internal heat and sweating during the practice helps cleanse the body.**
3. **Mental clarity and focus: The strong emphasis on breath and a fixed gaze helps calm the mind and improves concentration.**
4. **Stress reduction: The meditative aspect of the practice helps to reduce stress and anxiety, promoting inner peace.**
5. **Improved cardiovascular health: The flowing movements and consistent pace improve blood circulation and cardiovascular fitness.**
6. Improves mental concentration
7. Promotes moral values
8. Leads to spiritual development

- **PHYSICAL ADVANTAGES**

- **Increases strength and flexibility:** The challenging, flowing sequences build and tone muscles, particularly the core, legs, and arms, while also increasing your range of motion.
- **Improves cardiovascular health:** The vigorous, cardio-focused practice increases your heart rate and improves blood circulation, which helps to lower blood pressure.
- **Aids in weight management:** The intense, calorie-burning nature of the practice can help reduce body fat and contributes to a lean, toned physique.
- **Boosts energy and stamina:** The combination of physical endurance and controlled breathing increases overall energy and vitality.
- **Detoxifies the body:** The internal heat generated by the breath (ujjayi) and dynamic movement promotes sweating, which helps to cleanse and purify the body.
- **Enhances bone density:** As a weight-bearing exercise, regular practice can improve bone density and prevent age-related bone loss.
- **Supports spinal health:** Ashtanga can improve trunk and hamstring flexibility, which benefits spine health and can help alleviate back pain.
- **MENTAL AND EMOTIONAL ADVANTAGES**
- **Reduces stress and anxiety:** The focus required on the breath, gaze, and posture acts as a moving meditation that calms the nervous system and stimulates the brain's pleasure centers.
- **Increases mental clarity and focus:** The structured, repetitive sequences train the mind to concentrate, which can lead to improved mental focus and productivity in daily life.
- **Promotes emotional balance:** The practice can help release stored emotional tension and fosters emotional flexibility and resilience.
- **Develops self-awareness:** The deep inward focus required during practice creates a stronger mind-body connection and fosters a heightened sense of self-awareness.
- **Builds discipline and self-control:** The demanding and consistent nature of the practice encourages self-discipline, which can transfer to other areas of your life.
- **Enhances mood and well-being:** The combination of physical exercise and mindful practice can lead to a more optimistic outlook, improved self-confidence, and a general sense of well-being
- **Social-Community and relationship building, Community and relationship building**

ASHTANGA YOGA RELEVANCE IN TODAY CONTEXT

Ashtanga yoga is highly relevant in today's context as it offers a comprehensive framework for addressing modern challenges like chronic stress, physical inactivity, and the search for meaning in a materialistic world. Its integrated system of physical, mental, and ethical practices provides a path to holistic well-being and self-discovery.

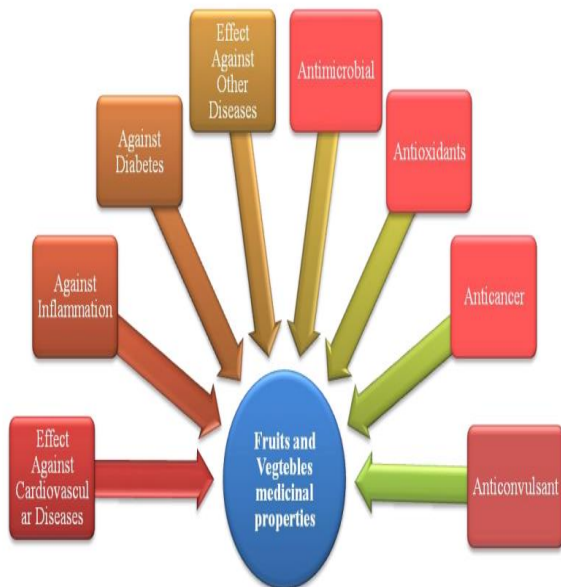
Addressing Modern Needs

- **Stress Management and Mental Clarity:** The fast-paced, demanding nature of modern life often leads to high levels of stress, anxiety, and mental health issues. Ashtanga's emphasis on breath control (**Pranayama**) and fixed gaze (**Drishti**) creates a moving meditation that helps to calm the nervous system, reduce racing thoughts, and improve concentration and focus.
- **Physical Health and Fitness:** In an age of sedentary lifestyles and physical decline, the rigorous and dynamic nature of the Ashtanga asana (posture) sequences builds significant strength, flexibility, stamina, and cardiovascular fitness. This physical discipline helps to manage weight, improve posture, and prevent chronic diseases.
- **Ethical Guidance and Self-Discipline:** The first two limbs, **Yamas** (ethical restraints) and **Niyamas** (personal observances), offer a timeless moral compass for navigating complex social interactions and personal challenges. Principles like *ahimsa* (non-violence), *satya* (truthfulness), and *santosha* (contentment) help individuals develop a sense of purpose, self-awareness, and harmonious relationships in a world often characterized by competition and materialism.
- **Mind-Body Connection:** The practice fosters a profound awareness of the connection between the body and mind, helping practitioners to process emotions, work through challenges, and develop emotional resilience.
- **Developing Discipline and Resilience:** The requirement for consistent, daily practice (traditionally six days a week) fosters self-discipline, perseverance, and emotional resilience. This "tapas" (inner fire/discipline) translates into a more structured approach to personal and professional responsibilities.
- **Mind-Body Connection and Self-Awareness:** In a world of digital distraction, Ashtanga's inward focus (pratyahara) helps practitioners tune into internal sensations, fostering a deep mind-body connection and heightened self-awareness. This self-study (svadhyaya) aids in understanding one's own patterns and emotional landscape.

- **Ethical Living:** The philosophical underpinnings of the eight limbs, particularly the *Yamas* (ethical principles like non-violence and truthfulness) and *Niyamas* (personal observances like contentment and cleanliness), provide a moral compass for conscious and purposeful living.
- **Community and Support:** Modern yoga studios and online platforms foster a strong global community (*sangha*) of like-minded individuals, offering a vital support system and sense of belonging in an often isolating world.
- **Adaptability for All Levels:** While traditionally considered rigorous, modern Ashtanga is increasingly adaptable, with teachers offering modifications and props for all ages, body types, and fitness levels, making the practice accessible to a wider audience.

MEDICINAL VALUES OF FRUITS

- Fruits are an indispensable part of any diet, not only for vegetarians, but also for nonvegetarians.
- Fruits are important sources of vitamins and minerals for the body.
- Fruits have many medicinal values, including:
- **Vitamin C** ; A water-soluble vitamin that acts as an antioxidant to maintain health. Citrus fruits, strawberries, and kiwifruit are high in vitamin C.
- **Fiber** ;Helps with gut health, metabolism, and weight loss. Dried fruits contain as much fiber as fresh fruits.
- **Potassium**;-An essential mineral that helps with muscle function and blood pressure regulation.
- **Antioxidants**;-Neutralize harmful free radicals, protect the body from oxidative stress, and reduce the risk of chronic diseases.
- **Folate**;- An essential micronutrient for DNA synthesis, repair, and methylation.
- **Calcium**;-A major building block for bone strength and overall health.
- **Vitamin A**;-A fat-soluble vitamin that acts as an anti-inflammatory and enhances immune function.
- **Immunity**;-Fruits are full of vitamin C, fiber, antioxidants, minerals, and enzymes that help produce white blood cells and fight infections.
- **Fruits are an indispensable part of any diet, not only for vegetarians, but also for nonvegetarians.**
- **Fruits are important sources of vitamins and minerals for the body.**
- Fruits are a good source of nutrients that can help with many aspects of health, including:
- **Digestion** ;Fiber in fruits helps with digestion, prevents constipation, and promotes a healthy gut.
- **Heart health**; Fruits can help reduce the risk of heart disease. For example, a diet rich in fruits, vegetables, and low-fat dairy can help lower blood pressure.
- **Weight management** ;-Fruits are low in calories and high in volume, which can help you feel full and satisfied.
- **Immune system** ;Fruits contain vitamins A and C, which can help strengthen the immune system.
- **Chronic disease risk** ;-Antioxidants and phytochemicals in fruits can reduce inflammation and lower the risk of chronic diseases like cancer and diabetes.
- **Blood pressure** ;Potassium in fruits can help maintain healthy blood pressure.
- **Tissue growth and repair** ;Vitamin C in fruits helps with the growth and repair of body tissues, and helps heal cuts and wounds.



Nutrient	Fruits
Moisture (g)	75-90
Calories (Kcal)	10-80
Carbohydrates (g)	2-20
Protein (g)	0.2-2.0
Fat (g)	0-1.0
Calcium (mg)	5-40
Iron (mg)	0.1-1.0
Carotene (μg)	5-500
Ascorbic acid (mg)	2-300
Thiamine (mg)	0.05-0.2
Riboflavin (mg)	0.02-0.1
Nicotini acid (mg)	0.2-1.0
Folic acid (μg)	6-9
Potassium (mg)	86-250
Sodium (mg)	5-15



Nutritional Value of Strawberries

FIBER:

regulate digestion
reduce the risk of cardiovascular and intestinal diseases

VITAMIN C:

protecting cells from damaging
keeps teeth and gums healthy
heals cuts and wounds
strengthens the immune system regulate digestion

POTASSIUM:

Regulate excretion

FOLIC ACID:

Helps your body to make red blood cells

Minerals:

Potassium - 220 mg
Phosphorus - 35 mg
Magnesium - 19 mg
Calcium - 23 mg
Sodium - 1 mg
Iron - 0.59 mg
Selenium 0.6 mcg
Manganese - 0.556 mg
Copper - 0.069 mg
Zinc - 0.2 mg

Vitamins:

Vitamin A - 17 IU
Vitamin B1 (thiamine) - 0.035 mg
Vitamin B2 (riboflavin) - 0.032 mg
Niacin - 0.556 mg
Folate - 35 mcg
Pantothenic Acid - 0.18 mg
Vitamin B6 - 0.068 mg
Vitamin C - 84.7 mg
Vitamin E - 0.42 mg
Vitamin K - 3.2 mcg

Health Benefits of Eating Strawberry

- Strawberries boost immunity
- Promote eye health
- Fight bad cholesterol
- Regulate blood pressure
- Aid in weight management



12 Amazing Benefits of STRAWBERRY

- Anti-oxidant
- Anti-inflammatory
- Weight loss
- Slows down skin aging
- Aids healthy digestion
- Improves skin health
- Promotes bone health
- Boosts immunity
- Promotes eye health
- Protects the heart
- Increases good cholesterol
- Lowers blood pressure

VITAMINS AND MINERALS IN BANANA:-

Vitamin B6	25%
Vitamin C	14%
Potassium	12%
Fiber	12%
Copper	10%
Biotin	10%
Manganese	16%

Nutritional Value of a Banana

Minerals:

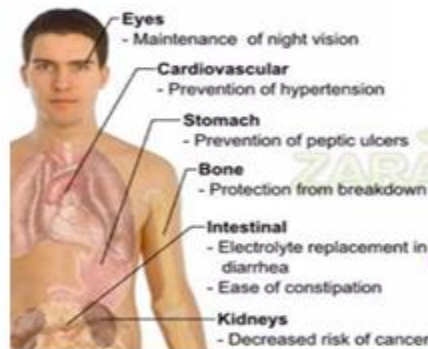
Potassium - 422 mg
Phosphorus - 26 mg
Magnesium - 32 mg
Calcium - 6 mg
Sodium - 1 mg
Iron - 0.31 mg
Selenium 1.2 mcg
Manganese - 0.319 mg
Copper - 0.092 mg
Zinc - 0.18 mg

Vitamins:

Vitamin A - 76 IU
Vitamin B1 (thiamine) - 0.037 mg
Vitamin B2 (riboflavin) - 0.086 mg
Niacin - 0.785 mg
Folate - 24 mcg
Pantothenic Acid - 0.394 mg
Vitamin B6 - 0.433 mg
Vitamin C - 10.3 mg
Vitamin E - 0.12 mg
Vitamin K - 0.6 mcg



HEALTH BENEFITS OF BANANAS



Other benefits:

- Restores healthy Blood Glucose level.
 - Rich in Iron.
 - Provides Energy.
 - Soothes Ulcers.
 - Increases Brain Power.



COMPOSITION

calories	62%
vitamin C	93%
fiber	13%
potassium	7%
calcium	5%
protein	1%
carbohydrate	6%
vitamin	2%

Nutritional Value of Oranges

Minerals:

Potassium - 237 mg
Phosphorus - 18 mg
Magnesium - 13 mg
Calcium - 52 mg
Iron - 0.13 mg
Selenium 0.7 mcg
Manganese - 0.033 mg
Copper - 0.059 mg
Zinc - 0.09 mg

Vitamins:

Vitamin A - 295 IU
Vitamin B1 (thiamine) - 0.114 mg
Vitamin B2 (riboflavin) - 0.052 mg
Niacin - 0.369 mg
Folate - 39 mcg
Pantothenic Acid - 0.328 mg
Vitamin B6 - 0.079 mg
Vitamin C - 69.7 mg
Vitamin E - 0.24 mg



Heart Health

Herperidin's in oranges help lower blood pressure and folate protects against cardiovascular disease



Digestive Health

Vitamin C helps prevent ulcers and fibre ensures a healthy colon



Cholesterol-Lowering

Contain limonin, which helps reduce LDL, or "bad" cholesterol



Kidney Support

Help prevent kidney stones & efficient filtering of toxins



Anti-Cancer

Studies have shown cancer-risk reductions in over 40-50% of individuals who consumed citrus fruits! Contain potent anti-carcinogens to prevent proliferation of cancerous cells.



Immune Support

High vitamin C content helps to steer away nasty bugs, bacteria & viruses. Prevents colds, flus & ear infections



Alkalizing

Rich in alkaline minerals to help balance body pH



Healthy Skin

Contain essential vitamins & minerals for beautiful, problem-free skin



Anti-Inflammatory

Oranges help prevent free-radical damage, which normally triggers the inflammatory cascade



Vision Protection

Loaded with carotenoids, oranges help prevent night blindness & macular degeneration



Nutritional Value of an Apple

Minerals:

Potassium - 195 mg
Calcium - 11 mg
Phosphorus - 20 mg
Magnesium - 9 mg
Manganese - 0.064 mg
Iron - 0.22 mg
Sodium - 2 mg
Copper - 0.049 mg
Zinc - 0.07 mg

Vitamins:

Vitamin A - 98 IU
Vitamin B1 (thiamine) - 0.031 mg
Vitamin B2 (riboflavin) - 0.047 mg
Niacin - 0.166 mg
Folate - 5 mcg
Pantothenic Acid - 0.111 mg
Vitamin B6 - 0.075 mg
Vitamin C - 8.4 mg
Vitamin E - 0.33 mg
Vitamin K - 4 mcg

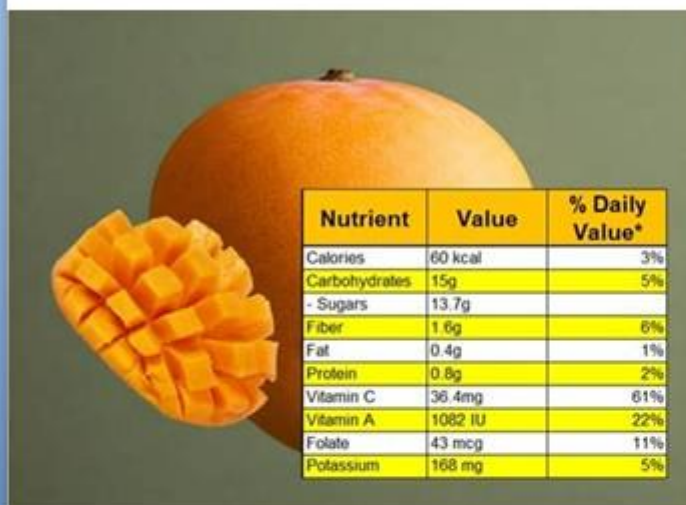
12 Incredible Health Benefits of Apples!

1. Stimulate Weight Loss
2. Balance Blood Sugar Levels
3. Strengthen Immunity
4. Improve Your Eyesight
5. Great for Your Teeth and Gums
6. Build Stronger Bones
7. Give You Energy
8. Relieve Muscle Tension
9. Improve Cardiovascular Health
10. Improve Fertility in Women
11. Improve Your Brain Function
12. Incredible Alkalizing



Nutrition Facts

Serving Size 1 fruit		207 g
Amount Per Serving		
Calories 135	Calories from fat 5	
		% Daily Value*
Total Fat 1g		1%
Saturated Fat 0g		1%
Trans Fat 0g		
Cholesterol 0g		0%
Sodium 4mg		0%
Total Carbohydrate 35g		12%
Dietary Fiber 4g		15%
Sugars 31g		
Protein 1g		
Vitamin A		32%
Vitamin C		96%
Calcium		2%
Iron		1%



Nutrient	Value	% Daily Value*
Calories	60 kcal	3%
Carbohydrates	15g	5%
- Sugars	13.7g	
Fiber	1.6g	6%
Fat	0.4g	1%
Protein	0.8g	2%
Vitamin C	36.4mg	61%
Vitamin A	1082 IU	22%
Folate	43 mcg	11%
Potassium	168 mg	5%

Many different pharmacological activities, antioxidant, radioprotective, immunomodulatory, anti-allergic, anti-inflammatory, antitumor, antidiabetic, lipolytic, antibone resorption, monoamine oxidase-inhibiting, antimicrobial and antiparasitic, have been reported for mangiferin

Nutritional Value of Grapes

Minerals:

Potassium - 288 mg
Phosphorus - 30 mg
Magnesium - 11 mg
Calcium - 15 mg
Sodium - 3 mg
Iron - 0.54 mg
Selenium - 0.2 mcg
Manganese - 0.107 mg
Copper - 0.192 mg
Zinc - 0.11 mg

Vitamins:

Vitamin A - 100 IU
Vitamin B₁ (thiamine) - 0.104 mg
Vitamin B₂ (riboflavin) - 0.106 mg
Niacin - 0.284 mg
Folate - 3 mcg
Pantothenic Acid - 0.076 mg
Vitamin B₆ - 0.13 mg
Vitamin C - 16.3 mg
Vitamin E - 0.29 mg
Vitamin K - 22 mcg

Health Benefits of Eating Grapes

Controlling Blood Sugar

Improving Brain Function

Protecting against infections and inflammations

Maintaining a Healthy and youthful Skin

1. Antioxidant Activity
2. Anti-Inflammatory Activity
3. Gut Micro biota Modulation
4. Antiobesity Activity
5. Cardio protective Activity
6. Antidiabetic Activity
7. Hepato | protective Activity
8. Anticancer Activity

MEDICINAL VALUES OF VEGITABLE

vegetables are an indispensable part of any diet, not only for vegetarians, but also for nonvegetarians.

They are plant derivatives and generally are matured

1. ovaries,
2. fruits,
3. flowers,
4. stems and
5. leaves of plants.

vegetables are important sources of vitamins and minerals for the body. Vegetables play a vital role in human nutrition and health by providing nutrients, vitamins, antioxidants, phytosterols, and dietary fiber.

Eating vegetables can also help with other health conditions, such as:

1. **Digestive health;**- Vegetables are a good source of dietary fiber, which can help keep bowel movements regular and prevent constipation.
2. **Blood pressure;**-Green leafy vegetables contain potassium, which can help reduce blood pressure.
3. **Heart health;** Green leafy vegetables contain vitamin K, which can help prevent calcium from building up in your arteries.
4. **Blood sugar management;**-Non-starchy vegetables have a low glycemic index, which means they have little effect on your blood sugar.
5. **Weight Management:** Vegetables are typically low in calories and fat but high in fiber, making them filling and helpful for managing weight
6. **Brain Health:** Certain vegetables, like leafy greens, are packed with antioxidants and folate, which may support brain function and potentially reduce the risk of cognitive decline
7. **Type 2 Diabetes:** A diet rich in vegetables can help improve blood sugar control, which can be beneficial for preventing or managing type 2 diabetes.
8. **Stroke:** Consuming a variety of vegetables has been linked to a reduced risk of stroke.
9. **Supporting Immune Function:** Vegetables are rich in vitamins and minerals, particularly vitamin C and beta-carotene, which are known to boost the immune system and help the body fight off infection

10. **Preventing Chronic Diseases:** A diet rich in vegetables can help reduce the risk of various chronic diseases, including heart disease, stroke, certain types of cancer, and type 2 diabetes
11. **Eye Health:** Some vegetables, like carrots, are rich in vitamin A, which is essential for good vision.
12. **Bone Health:** Leafy greens like kale and spinach are good sources of vitamin K, which is important for bone health and blood clotting.
13. **Skin Health:** Some vegetables, like tomatoes, contain antioxidants that can help protect the skin from damage caused by the sun.
14. **Weight Management:** Vegetables are generally low in calories and fat, making them a good choice for weight management

NUTRITION INFORMATION FOR RAW VEGETABLES

Vegetables Serving Size (gram weight/ ounce weight)	Calories	Calories from Fat	Total Fat		Sodium		Potassium		Total Carbo- hydrate		Dietary Fiber		Sugars	Protein	Vitamin A (%DV)	Vitamin C (%DV)	Calcium (%DV)	Iron (%DV)
			(g)	(%DV)	(mg)	(%DV)	(mg)	(%DV)	(g)	(%DV)	(g)	(%DV)						
Asparagus 5 spears (93 g/3.3 oz)	20	0	0	0	0	0	230	7	4	1	2	8	2	2	10	15	2	2
Bell Pepper 1 medium (148 g/5.3 oz)	25	0	0	0	40	2	220	6	6	2	2	8	4	1	4	190	2	4
Broccoli 1 medium stalk (148 g/5.3 oz)	45	0	0.5	1	80	3	460	13	8	3	3	12	2	4	6	220	6	6
Carrot 1 carrot, 7" long, 1 1/4" diameter (78 g/2.8 oz)	30	0	0	0	60	3	250	7	7	2	2	8	5	1	110	10	2	2
Cauliflower 1/6 medium head (99 g/3.5 oz)	25	0	0	0	30	1	270	8	5	2	2	8	2	2	0	100	2	2
Celery 2 medium stalks (110 g/3.9 oz)	15	0	0	0	115	5	260	7	4	1	2	8	2	0	10	15	4	2
Cucumber 1/3 medium (99 g/3.5 oz)	10	0	0	0	0	0	140	4	2	1	1	4	1	1	4	10	2	2
Green (Snap) Beans 3/4 cup cut (83 g/3.0 oz)	20	0	0	0	0	0	200	6	5	2	3	12	2	1	4	10	4	2
Green Cabbage 1/12 medium head (84 g/3.0 oz)	25	0	0	0	20	1	190	5	5	2	2	8	3	1	0	70	4	2

Nutritional Value of Potatoes

Minerals:

Potassium - 926 mg
Phosphorus - 121 mg
Magnesium - 48 mg
Calcium - 26 mg
Iron - 1.87 mg
Sodium - 17 mg
Zinc - 0.62 mg
Copper - 0.204 mg
Manganese - 0.379 mg
Selenium - 0.7 mcg

Vitamins:

Vitamin C - 16.6 mg
Niacin - 2.439 mg
Vitamin B₁ (thiamine) - 0.111 mg
Vitamin B₂ (riboflavin) - 0.083 mg
Pantothenic Acid - 0.65 mg
Vitamin B₆ - 0.538 mg
Folate - 48 mcg
Vitamin A - 17 IU
Vitamin K - 3.5 mcg
Vitamin E - 0.07 mg



Health Benefits of Eating Potato

- Ability to improve digestion
- Reduce cholesterol levels
- Boost heart health
- Prevent cancer
- Manage diabetes
- Strengthen the immune system



Nutritional Value of Carrots

- **Minerals:**
 - **Potassium** - 183 mg
 - **Calcium** - 23 mg
 - **Phosphorus** - 23 mg
 - **Magnesium** - 8 mg
 - **Iron** - 0.27 mg
 - **Sodium** - 5 mg
 - **Zinc** - 0.3 mg
 - **Copper** - 0.052 mg
 - **Manganese** - 0.062 mg
 - **Selenium** - 0.2 mcg
- **Vitamins:**
 - **Vitamin A** - 13286 IU
 - **Vitamin C** - 2.8 mg
 - **Vitamin B₁** (thiamine) - 0.051 mg
 - **Vitamin B₂** (riboflavin) - 0.034 mg
 - **Niacin** - 0.503 mg
 - **Folate** - 11 mcg
 - **Pantothenic Acid** - 0.181 mg
 - **Vitamin B₆** - 0.119 mg
 - **Vitamin K** - 10.7 mcg
 - **Vitamin E** - 0.8 mg



Health Benefits of Eating Carrots

- **Protects Eye Health**
- **High Source of Antioxidants (Especially Beta Carotene)**
- **Helps Protect Against Cancer**
- **Important for Maintaining Oral Health**
- **Boosts Skin Health and Wound Healing**
- **Decreases Risk for Heart Disease and Stroke**



Nutritional Value of Tomatoes

- **Minerals:**
 - **Potassium** - 292 mg
 - **Phosphorus** - 30 mg
 - **Magnesium** - 14 mg
 - **Calcium** - 12 mg
 - **Sodium** - 6 mg
 - **Iron** - 0.33 mg
 - **Manganese** - 0.14 mg
 - **Copper** - 0.073 mg
 - **Zinc** - 0.21 mg
- **Vitamins:**
 - **Vitamin A** - 1025 IU
 - **Vitamin B₁** (thiamine) - 0.046 mg
 - **Vitamin B₂** (riboflavin) - 0.023 mg
 - **Niacin** - 0.731 mg
 - **Folate** - 18 mcg
 - **Pantothenic Acid** - 0.109 mg
 - **Vitamin B₆** - 0.098 mg
 - **Vitamin C** - 15.6 mg
 - **Vitamin E** - 0.66 mg
 - **Vitamin K** - 9.7 mcg



Health Benefits of Eating Tomato

- **Tomatoes help maintain strong bones.**
- **Reduces Cholesterol and Protects the Heart.**
- **Lowers Hypertension.**
- **Protects vision and degenerative eye disease.**



Nutritional Value of Cucumber

- **Minerals:**
 - **Potassium** - 76 mg
 - **Phosphorus** - 12 mg
 - **Magnesium** - 7 mg
 - **Sodium** - 1 mg
 - **Calcium** - 8 mg
 - **Iron** - 0.15 mg
 - **Zinc** - 0.1 mg
 - **Copper** - 0.021 mg
 - **Manganese** - 0.041 mg
 - **Selenium** - 0.2 mcg
- **Vitamins:**
 - **Vitamin C** - 1.5 mg
 - **Niacin** - 0.051 mg
 - **Vitamin B₁** (thiamine) - 0.014 mg
 - **Vitamin B₂** (riboflavin) - 0.017 mg
 - **Vitamin B₆** - 0.021 mg
 - **Folate** - 4 mcg
 - **Pantothenic Acid** - 0.135 mg
 - **Vitamin A** - 55 IU
 - **Vitamin K** - 8.5 mcg
 - **Vitamin E** - 0.02 mg



Health Benefits of Eating Cucumber

- **Protect Your Brain**
- **Freshen Your Breath**
- **Manage Stress**
- **Support Your Digestive Health**
- **Maintain a Healthy Weight**
- **Rehydrates body and replenishes daily vitamins**
- **Protect your brain from neurological diseases**
- **Protect your skin against the effects of aging**



Nutritional Value of Long Beans

Minerals:

Potassium - 655 mg
Phosphorus - 181 mg
Magnesium - 99 mg
Calcium - 112 mg
Iron - 1.91 mg
Sodium - 11 mg
Zinc - 1.13 mg
Copper - 0.204 mg
Manganese - 0.676 mg
Selenium - 2.1 mcg

Vitamins:

Vitamin C - 2.1 mg
Niacin - 0.966 mg
Vitamin B₁ (thiamine) - 0.23 mg
Vitamin B₂ (riboflavin) - 0.11 mg
Vitamin B₆ - 0.186 mg
Folate - 133 mcg
Pantothenic Acid - 0.393 mg
Vitamin A - 5 IU



Health Benefits of Eating Long Beans

- Maintains cell and body fluid
- Good source of nutrients that help to prevent many infections in the body
- Increase stamina and overcome extreme tiredness



MEDICINAL VALUE OF SPICES

- **Types of Spices with Names**
- There are different categories of spices used in cooking, each contributing its own unique flavour and aroma to dishes.
- **1. Whole Spices**
 - Cumin Seeds (Jeera)
 - Coriander Seeds (Dhaniya)
 - Fennel Seeds (Saunf)
 - Black Mustard Seeds (Rai)
 - Fenugreek Seeds (Methi)
- **2. Ground Spices**
 - Turmeric Powder (Haldi)
 - Red Chili Powder (Lal Mirch)
 - Coriander Powder (Dhaniya Powder)
 - Cumin Powder (Jeera Powder)
 - Ginger Powder (Sonth)
- **3. Aromatic Spices**
 - Cardamom (Elaichi)
- **Cinnamon (Dalchini)**
- **Cloves (Laung)**
- **Nutmeg (Jaiphal)**
- **Mace (Javitri)**
- **4. Leaf Spices**
 - Bay Leaf (Tej Patta)
 - Curry Leaves (Kadi Patta)
 - Mint (Pudina)
 - Fenugreek Leaves (Kasuri Methi)
 - Coriander Leaves (Dhaniya Patta)
- **5. Exotic Spices**
 - Saffron (Kesar)
 - Star Anise (Chakri Phool)
 - Black Cardamom (Badi Elaichi)
 - Caraway Seeds (Shahi Jeera)
 - Dried Pomegranate Seeds (Anardana)

Spices have many medicinal values, including:

Antioxidants: Spices can increase oxygenation and circulation, and boost the immune system.

Anti-inflammatory: Some spices, like turmeric, can help prevent inflammatory diseases like digestive and joint pain.

Antimicrobial: Some spices, like cloves, can be used as an antimicrobial.

Pain relief: Some spices, like cardamom, can be used as a pain reliever.

Digestive aid: Some spices, like cardamom, can help with digestion.

Cardiovascular health: Some spices, like oregano, can promote cardiovascular health.

Gastrointestinal health: Some spices, like oregano, can promote gastrointestinal health.

Brain and heart health: Some spices, like rosemary, can boost brain and heart health.

Stress management: Some spices, like cumin, can help with stress management.

Weight loss: Some spices, like cumin, can help with weight loss.

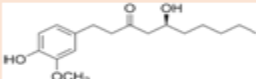
Cholesterol: Some spices, like cumin, can help with cholesterol.

Antidiabetic: Some spices, like cumin and fenugreek, can help with diabetes.

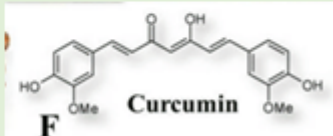
GARLIC (Source: Handbook of Herbs and Spices – Ch.17, U.B.Pandey, Jain Irrigation Systems Ltd)

BOTONICAL NAME ORGANIC COMPOUNDS <i>Allium Sativum</i>  $2\text{CH}_2=\text{CH}-\text{CH}_2-\text{S}-\text{CH}_2(\text{NH}_2)\text{COOH CH}_2=\text{CH}_2$ Alliin - Carbohydrates = 30% - Protein= 7% - Vitamic C= 13% - Moisture = 50% - Distinct smell of garlic is due to Sulphur content	PRESERVATIVE VALUE - Anti-oxidant - Anti-microbial - Used in Food Products - Used in Pest Control THERAPUTIC VALUE - Prevents Cancer - Reduces the blood pressure - Reduces cholesterol - Antibiotic – microbes, virus - Increases Immunity - Reduces mucus formation - Reduces uric acid - Potassium = 4 mg/g - Sulphur = 1 mg/g - Phosphorus = 1.5 mg/g - Magnesium = 0.3 mg/g - Calcium = 0.9 mg/g	CULTIVATION & VALUE - India Produces 30,00,000 tons = 30 lakh ton/year - Garlic is Cultivated in the following states : - Madhya Pradesh - Uttar Pradesh - Rajasthan - Gujarat - Price : Rs.150-250/- based on Varieties - India Exports = 60,000 tons = Rs.1000 crore
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GINGER (Source: Handbook of Herbs and Spices – Ch.18, P.A.Vasala, Kerala Agricultural Univ.)

BOTONICAL NAME ORGANIC COMPOUNDS <i>Zingiber Officinale</i> (Rhizome) Indian Name: Singabera   - Starch = 53% - Protein= 13% - Fiber = 7% - Oil = 3%	PRESERVATIVE VALUE - Anti-oxidant - Anti-microbial - Used in biscuits, bread, food products THERAPUTIC VALUE - Relieves spasm of muscles - Prevents gas formation - Prevents Vomiting , Nausea - Anti-inflammatory - Removes excess urea 1-4 gm a day is recommended - Potassium = 1.5mg/g - Sodium = 0.3 mg/g - Phosphorus = 1.5 mg/g	CULTIVATION & VALUE - India is second largest exporter of Ginger 250,000 Tons production per year - Average Price = Rs.140/kg - to Rs.250/kg – Variety / Climate and quality determines the price - Total Export = Rs.3500 crore/yr - Maharashtra - Assam - Meghalaya - West Bengal - Karnataka - Kerala 
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TURMERIC (Source: Handbook of Herbs and Spices – Ch.28, B.Sasikumar, Indian Institute of Spice Research)

BOTONICAL NAME ORGANIC COMPOUNDS Rhizome of the Plant <i>Curcuma Domestica</i>, <i>Curcuma Longa</i> is one of the 80 varieties. Indian Name: Haldi  - Carbohydrates = 53% - Olioresin= 15% (biopolymer) - Oil = 5% - Fibre = 4%	PRESERVATIVE VALUE - Anti-bacterial - Anti-microbial - Anti-oxidant Hence is used in food products to increase the shelf life, in addition to taste & flavour THERAPUTIC VALUE 4 kcal/gm - Potassium 20mg /g of turmeric - Phosphorous 4 mg/g of turmeric - Antiseptic paste for cuts, wounds - Good for digestion, to prevent respiratory ailments - Anti-inflammatory	CULTIVATION & VALUE - India is the largest exporter, producer, consumer of Turmeric 12 lakh Tons per year is produced - Average cost = Rs.200/kg - Total Export = Rs.1000 crore/yr, Middle East and America - Maharashtra - Karnataka - Tamil Nadu - Telangana & Andhra 
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MUSTARD (Source: Handbook of Herbs and Spices – Ch.21, J.Thomas, ICRA Spice Board.)

BOTONICAL NAME ORGANIC COMPOUNDS

Brassica nigra – Black Mustard
Brassica hirta – Yellow Mustard



- Protein = 25%
- Fat = 28 %
- Carbohydrates = 35%

THERAPUTIC VALUE

- Enhances digestion
- When applied on the skin reduces irritation
- Prevents gas formation
- Prevents Nausea
- Mustard oil + Camphor paste is used to treat cuts and wounds.
- Energy: 4 kCal/gm
- Potassium = 7mg/1g
- Phosphorus = 8 mg/g

Food Additive

- Mustard oil is used for cooking veg and non-veg dishes.
- Aroma and flavour of mustard is important for many people across the world.

CULTIVATION & VALUE

- India is exports mustard oil to USA, Canda and Europe
- 120 lakh ton per year
- Average Price = Rs.60/kg
- Total Export = Rs.15000 crore/yr
- Mustard Oil is widely used.
- Rajasthan
- Haryana
- Uttar Pradesh
- Gujarat

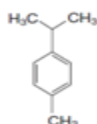
CUMIN (Source: Handbook of Herbs and Spices – Ch.13, G.Amin, Tehran University of Medical Science)

BOTONICAL NAME ORGANIC COMPOUNDS

Cuminum



- Carbohydrate 44%
- Protein= 17%
- Fiber = 10%
- Fat = 22%



PRESERVATIVE VALUE

- Anti-microbial
- Anti-Fungal
- Anti-Oxidant

THERAPUTIC VALUE

- Enhances digestion
- Prevents Cancer
- Reduces Blood pressure
- Energy: 4 kCal/gm
- Potassium = 18 mg/g
- Phosphorus = 5 mg/g
- Magnesium = 4 mg/g
- Sodium = 0.2 mg/g
- Iron = 0.6 mg/g

Food Additive

- Aroma and flavour of mustard is important for many people across the world.

CULTIVATION & VALUE

- Cultivation of Cumin is dominated by
- Gujarat
- Rajasthan
- Production: 7,00,000 Tons/year
- Export = 2,00,000 Tons/year
- Price per kg = Rs.400
- Total Exports = Rs.8000 crore per year



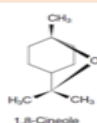
CARDAMOM (Source: Handbook of Herbs and Spices – Ch.8, V.A.Parthasarathy & D.Prasath, Indian Inst Spice Res.)

BOTONICAL NAME ORGANIC COMPOUNDS

Elettaria Cardamomum
Indian Name: "Ela" chi



- Carbohydrate 60%
- Protein= 10%
- Fiber = 30%



FOOD VALUE

- Aroma, Taste / Flavour in foods
- Bakery products
- Indian Sweet Dishes
- Beverage – Elaichai is common

THERAPUTIC VALUE

- Improves respiration
- Prevents gas formation
- Anti-inflammatory
- Promotes Digestion
- Sore throat
- Reduces exhaustion
- Honey + Elaichai
- Potassium = 11mg/g
- Magnesium = 3 mg/g
- Phosphorus = 2mg/g
- Calcium = 4mg/g

CULTIVATION & VALUE

- Major Producers are:
- Tamil Nadu
- Karnataka
- Kerala
- Price per Kg = Rs.1000 to Rs.4000
- Production: 25,000 Tons/year
- Export = 6000 Ton/year
- Middle East, Europe, Australia,
- America/Canada

